Action Based Testing (ABT) Cheat Sheet

The Three-Layer Architecture

Layer	Purpose	Changes When
Business Layer	WHAT to test (user goals)	Business requirements change
Interaction Layer	HOW to interact (reusable actions)	Workflow changes
Interface Layer	WHERE to click (UI specifics)	UI changes

Key Principles

1. Separation of Concerns

- Business logic stays in business layer
- UI details isolated in interface layer
- One change = one place to update

2. Test Module Structure

```
TEST MODULE: Login Validation
├── OBJECTIVE: Verify user authentication
├── PRECONDITIONS: User exists in system
├── TEST CASES:
│   ├── TC1: Valid credentials → Success
│   ├── TC2: Invalid password → Error message
│   └── TC3: Empty fields → Validation error
└── CLEANUP: Logout user
```

3. Naming Conventions

- **Actions**: verb + noun (e.g., `click button`, `enter text`)
- **Test Modules**: descriptive purpose (e.g., `validate login`)
- **Arguments**: clear, typed (e.g., `username: string`)

The 13 Anti-Patterns to Avoid

Anti-Pattern	Problem	Solution
Hardcoded Values	Brittle tests	Use variables/data files
Copy-Paste Tests	Maintenance nightmare	Create reusable modules
UI-Dependent Logic	Breaks on UI changes	Use three-layer separation
Missing Cleanup	Test pollution	Always include cleanup steps
Implicit Waits	Flaky tests	Use explicit conditions
Over-Assertion	Hard to debug	One concept per assertion
Test Interdependence	Cascade failures	Independent test modules
Magic Numbers	Unclear intent	Named constants
Commented Code	Confusion	Delete or document
Giant Test Cases	Unmaintainable	Break into modules
No Error Handling	Silent failures	Explicit error checks
Environment Coupling	Can’t run anywhere	Parameterize environment
Missing Documentation	Knowledge silos	Document purpose & usage

Quick Reference: Test Life-Cycle

```
1. SETUP → Prepare test environment
2. EXECUTE → Run test actions
3. VERIFY → Check expected results
4. CLEANUP → Restore initial state
5. REPORT → Log results
```

Agile Integration

- **Sprint Planning**: Identify testable user stories
- **During Sprint**: Build test modules alongside features
- **Sprint Review**: Demo automated tests
- **Cross-over Testing**: Test interactions between sprint items and existing features

Decision Table Template

Condition 1	Condition 2	Condition 3	Expected Result
True	True	True	Action A
True	True	False	Action B
True	False	*	Action C
False	*	*	Action D

Remember

> *"The best test is one that finds bugs, survives changes, and explains itself."*

Generated from Ask Marilyn About Software Testing - ABT Fundamentals Course